Monitoring global trends in Covid-19 vaccination intention and confidence: a social media-based deep learning study

挑战杯答辩

指导老师：公共卫生学院 侯志远副教授
答辩人：周新宇 17级本科

Introduction

Background

新冠疫情泛滥，控制疫情的希望在于疫苗接种，各国接种意愿很重要

GAP

1）大多数调查局限在少数国家，迄今最大的全球新冠疫苗犹豫研究仅比较了32国，许多国家被现有研究忽视；（WHO推荐各国根据疫苗犹豫状况等国情制定差异化的接种策略）

2）现有研究表明疫苗犹豫随时间有较大变化，但世界范围内各国的时间变化趋势未被研究

Aim

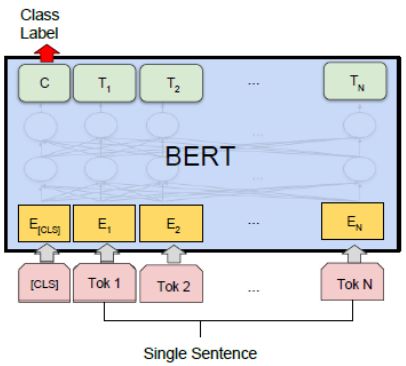
1) 100+国家新冠疫苗犹豫（及其影响因素）的流行状况与时间趋势

2) 各国家、区域新冠疫苗犹豫潜在诱因/对策

Methodology

1）爬取约700万条新冠疫苗相关推特（全样本）
2）利用课题组背靠背人工标记的~8000条推特训练基于CT-BERT的深度学习模型，(多标签)分类推特文本
3）用各标签在（特定时间、地点的）全部推特的占比来评估公众态度

About CT-BERT



(b) Single Sentence Classification Tasks: SST-2, CoLA

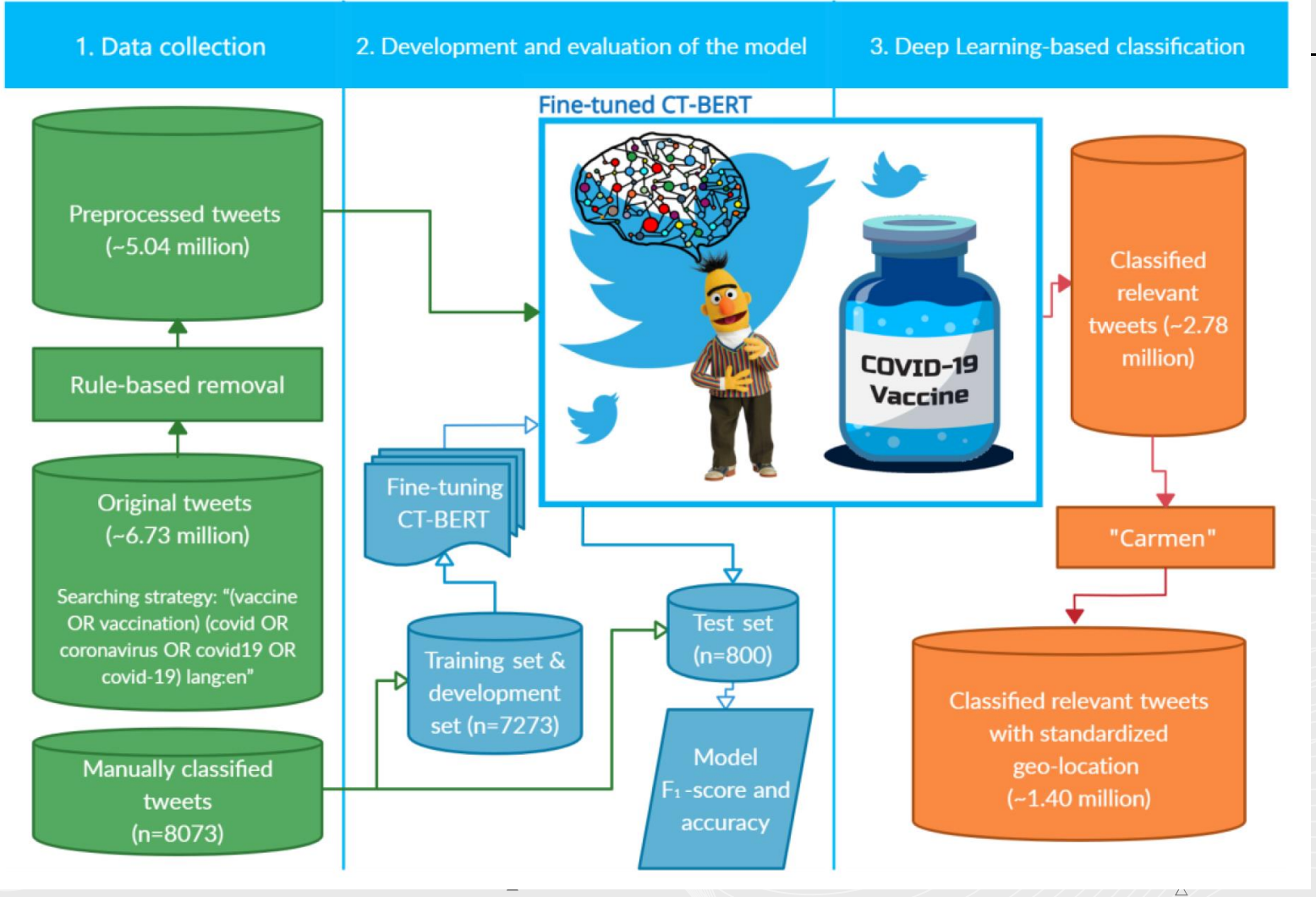
System	Dev	Test
ESIM+GloVe	51.9	52.7
ESIM+ELMo	59.1	59.2
OpenAI GPT	-	78.0
$BERT_{BASE}$	81.6	-
$BERT_{LARGE}$	**86.6**	**86.3**
Human (expert)†	-	85.0
Human (5 annotations)†	-	88.0

COVID-Twitter-BERT (CT-BERT)的原型是Bidirectional Encoder Representations from Transformers(BERT) (2018, 16000+ citation).

BERT是采用了Transformers (2017, 18000+citation)注意力机制的12/24层神经网络，并在数十亿语料中预训练，使得其在General Language Understanding Evaluation benchmark所有测试项目中一致超越先前所有自然语言处理模型，相比之前最好的模型提升了7%的绝对精度

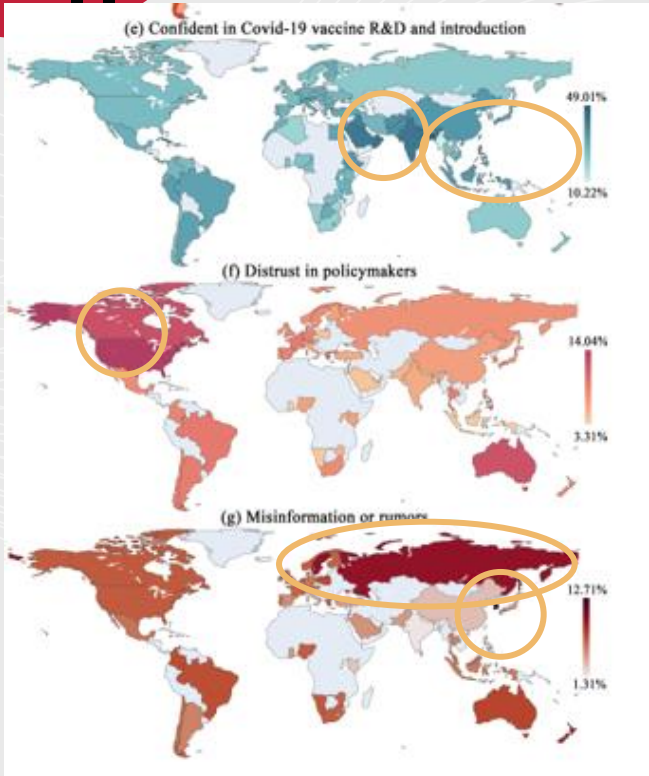
CT-BERT (2020) 采用了与BERT相似的预训练机制，进一步用新冠相关的1.6亿条推特预训练24层版BERT，使其成为最适合本研究的模型之一。

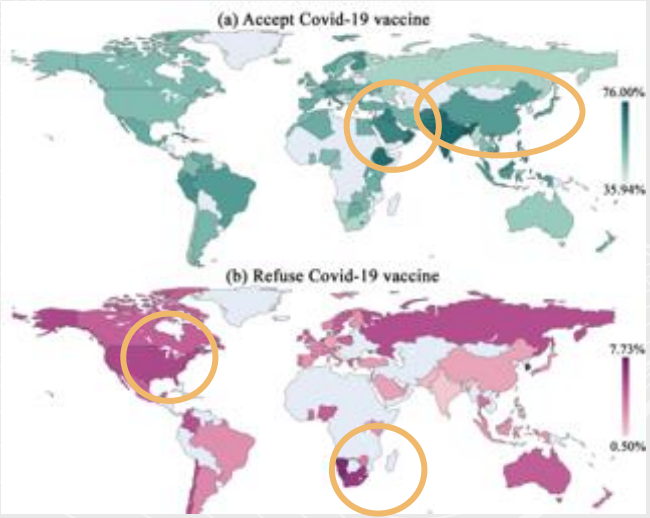
Ashish Vaswani et al. (2017); Jacob Devlin et al.(2018); Martin Müller et al. (2020)

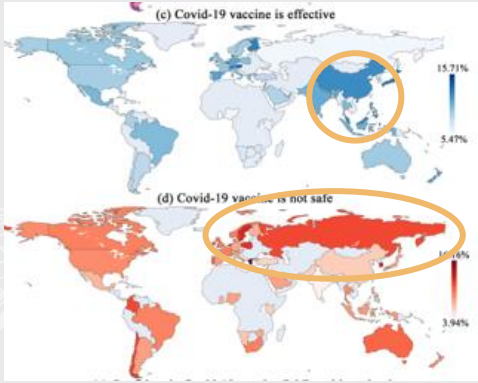

1. Data collection
2. Development and evaluation of the model
3. Deep Learning-based classification
Fine-tuned CT-BERT
Preprocessed tweets (~5.04 million)
Rule-based removal
Original tweets (~6.73 million)
Searching strategy: "(vaccine OR vaccination) (covid OR coronavirus OR covid19 OR covid-19) lang:en"
Manually classified tweets (n=8073)
Fine-tuning CT-BERT
Training set & development set (n=7273)
Test set (n=800)
Model F1-score and accuracy
COVID-19 Vaccine
Classified relevant tweets (~2.78 million)
"Carmen"
Classified relevant tweets with standardized geo-location (~1.40 million)

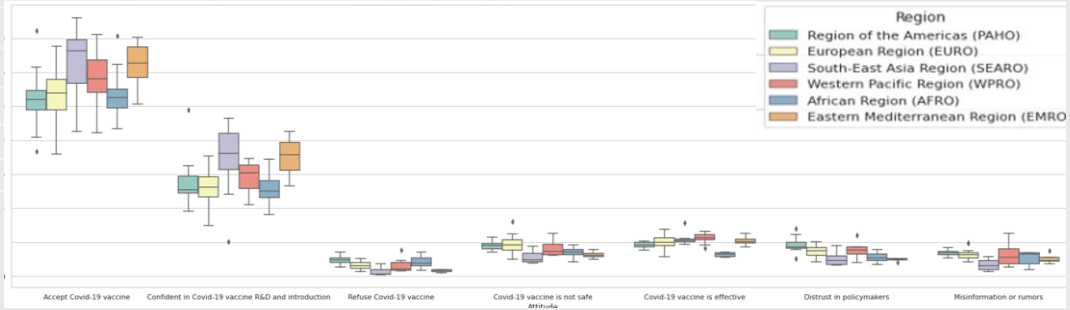
Results

全球新冠疫苗犹豫/影响因素分布



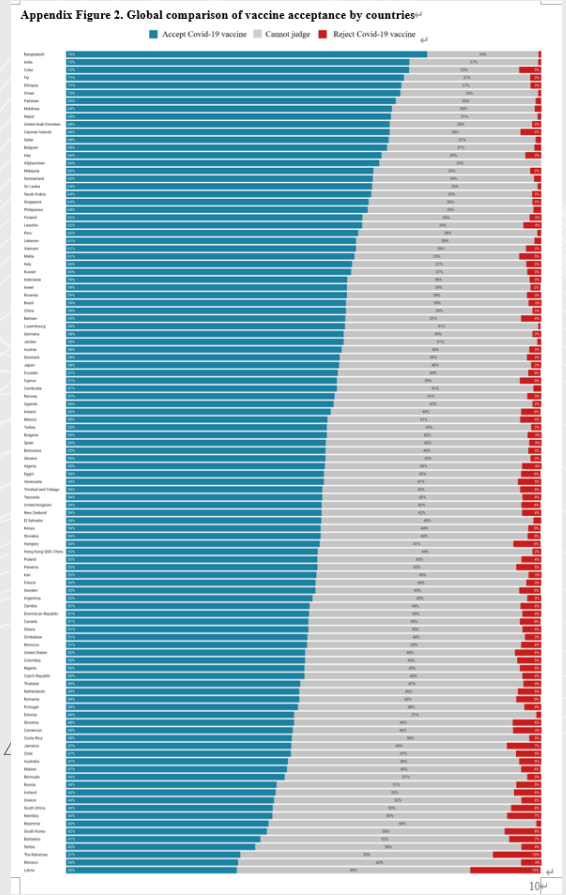


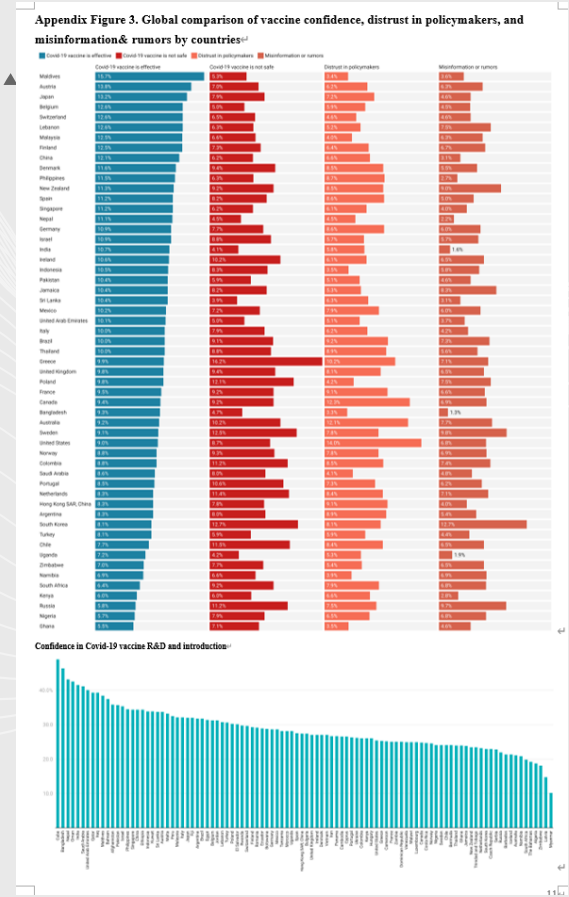




Results

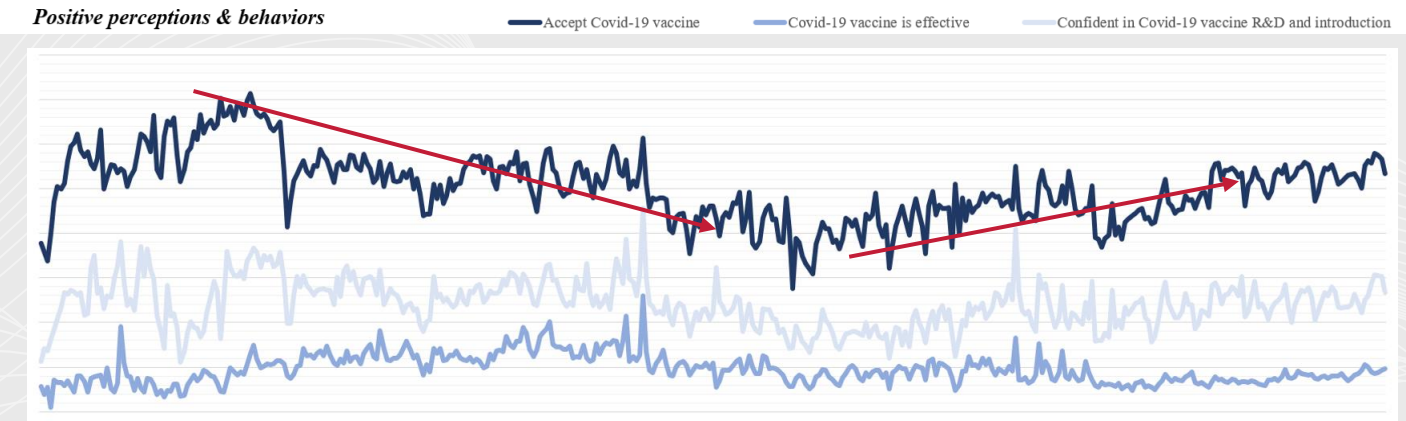
全球新冠疫苗犹豫/影响因素分布（附录·柱状图）

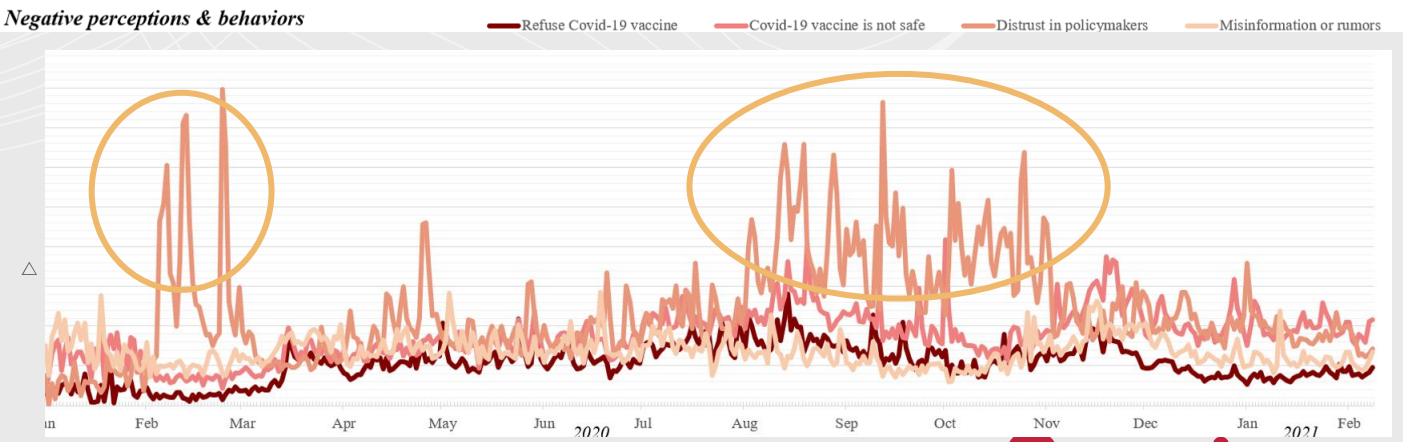




Results

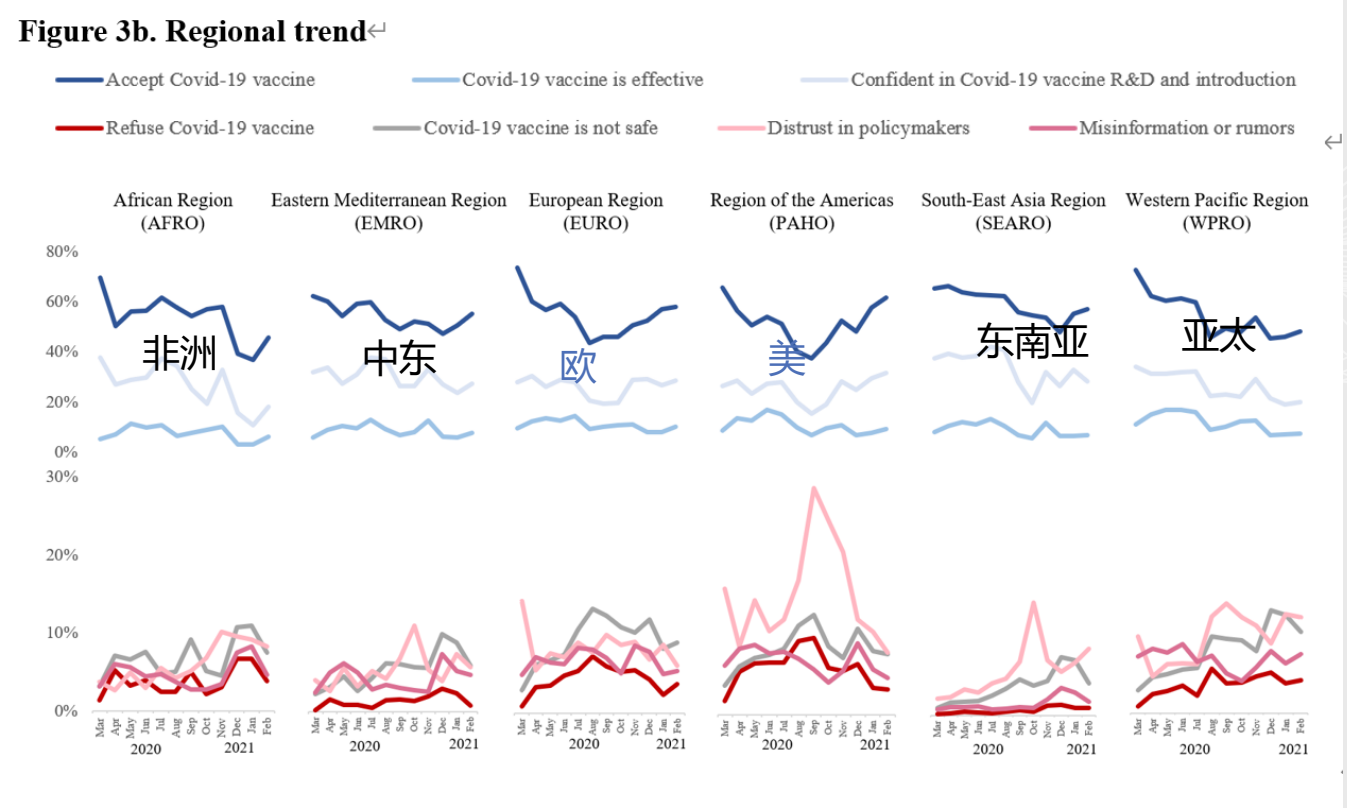
全球新冠疫苗接受度变化





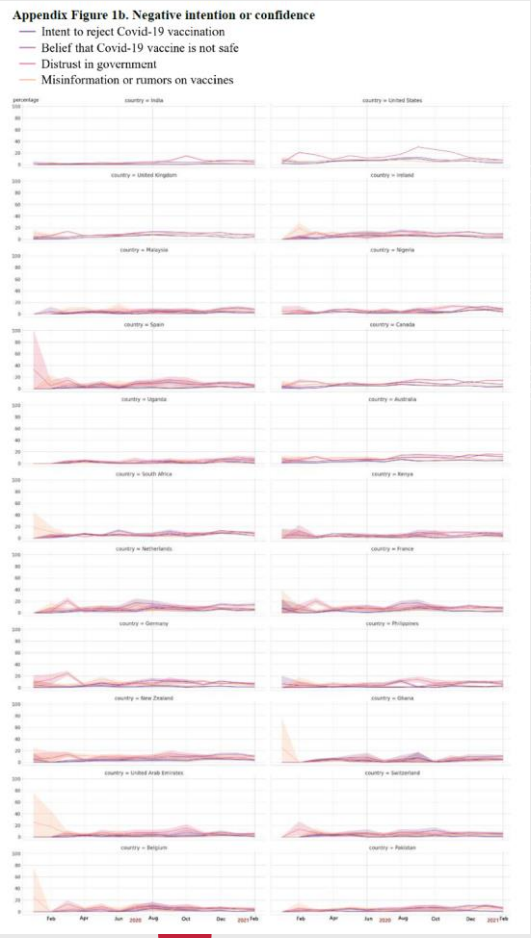
Results

全球（按区域）新冠疫苗接受度变化



Results

全球（各国）新冠疫苗接受度



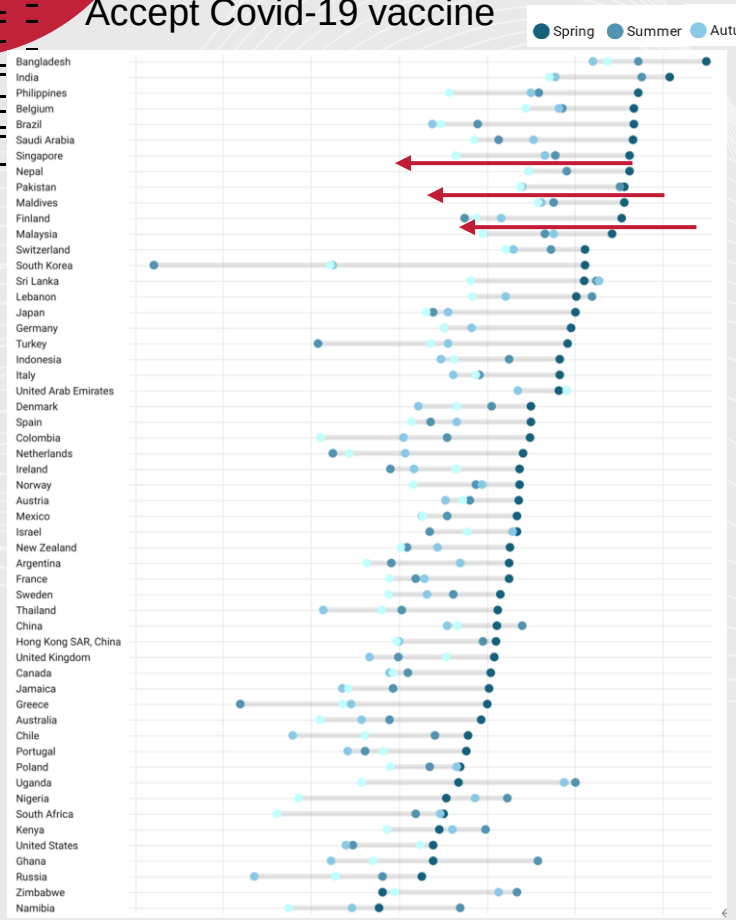
Results

全球新冠疫苗犹豫/影响因素按季节变化

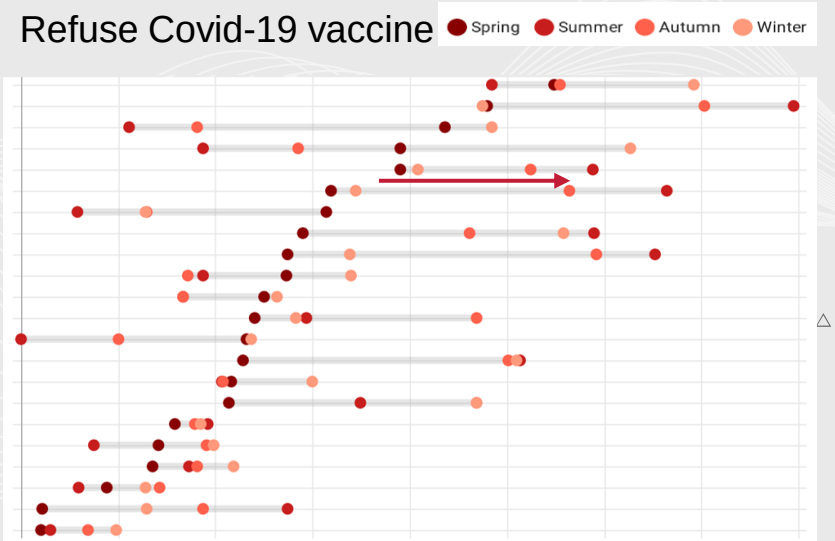
Spring：2020.3-5，Summer：2020.6-8，Autumn：2020.9-11，Winter：2020.12-2021.2

深色→2020年初，浅色→今年年初

Accept Covid-19 vaccine



Refuse Covid-19 vaccine



疫苗信心

Figure 5c. Covid-19 vaccine is effective

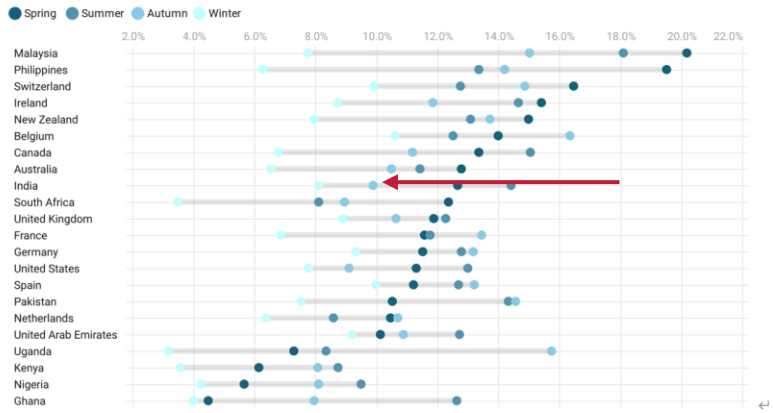


Figure 5d. Covid-19 vaccine is not safe

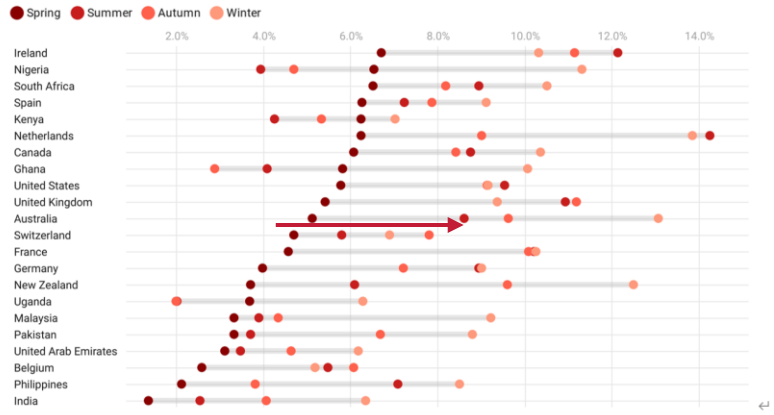
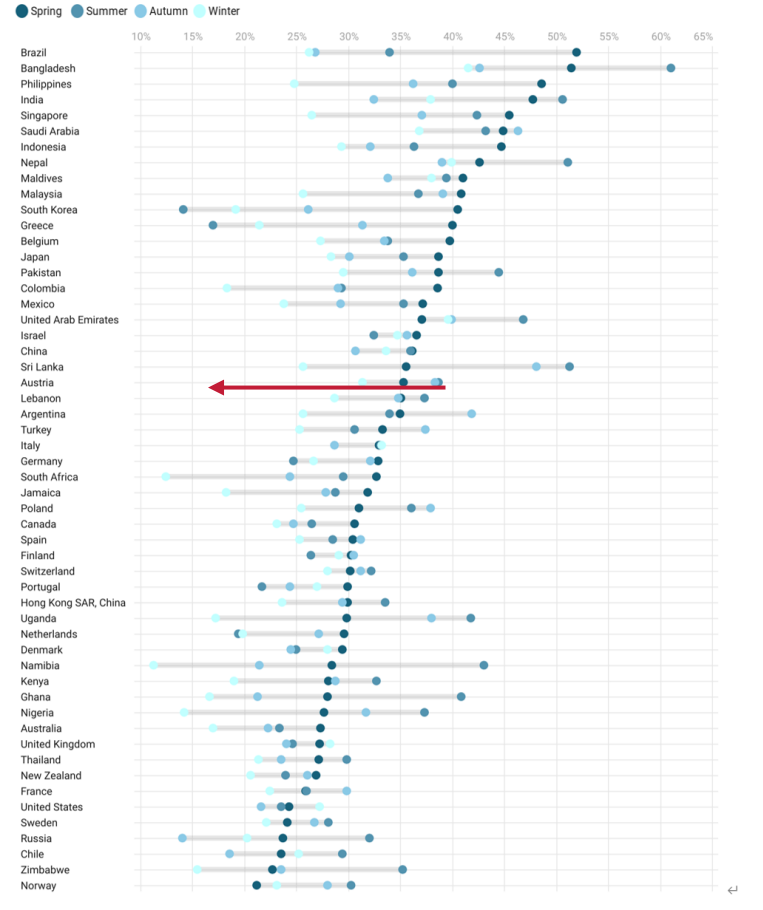
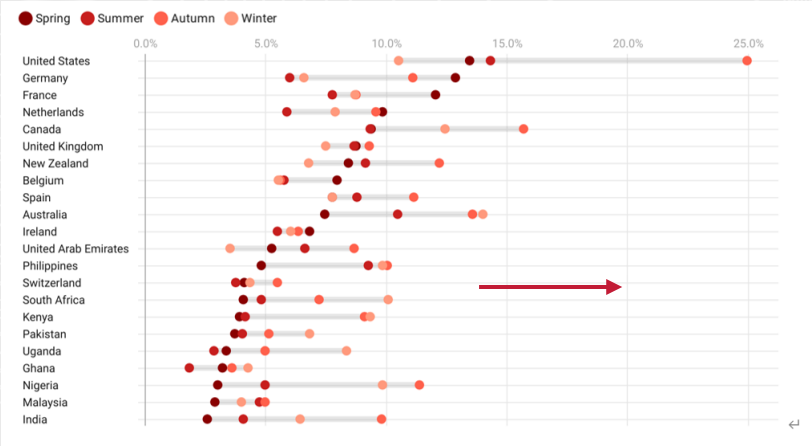


Figure 5e. Confident in Covid-19 vaccine R&D and introduction

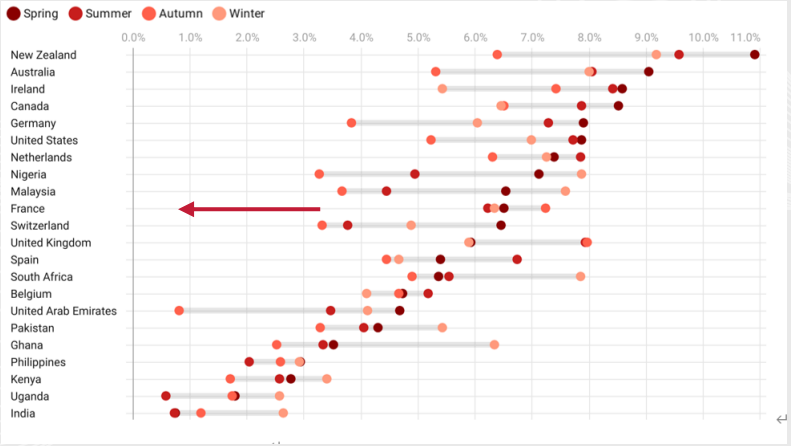


谣言/政府信任度

不信任政府



谣言



Results

SLR：

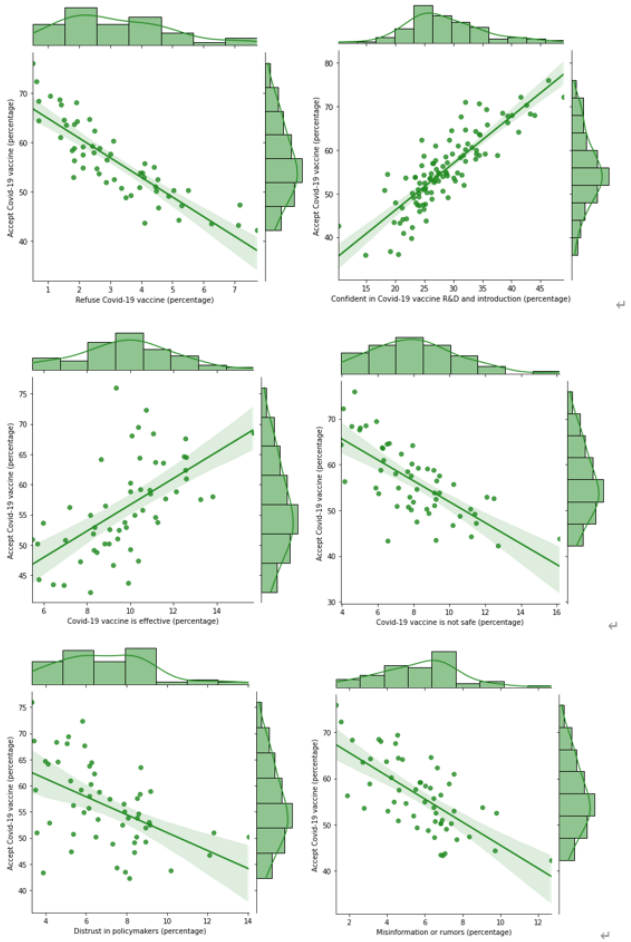
内部因素：强相关(p=0.000)
疫苗不安全、谣言、不信任政府的国家→疫苗犹豫

认为疫苗有效、对疫苗上市研发有信心的国家→支持疫苗

外部因素：Aging强相关(p=0.006, 负相关)
→老龄化严重的发达国家支持率低

新冠死亡人口百分比负相关（P=0.09)→不支持新冠疫苗的国家，可能还盛行其他增加新冠风险的行为

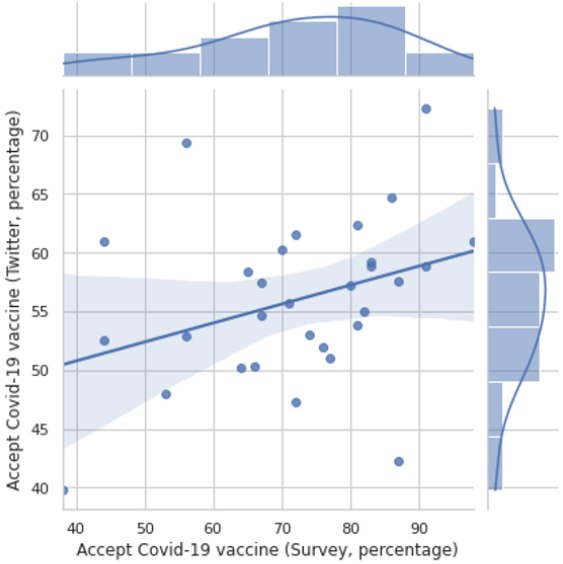
Appendix Figure 5. Country-level correlation between accepting Covid-19 vaccine and influencing factors



Results

稳健性——与现有问卷研究的比较

Appendix Figure 6. Comparison of social media data in our study and the largest survey published elsewhere



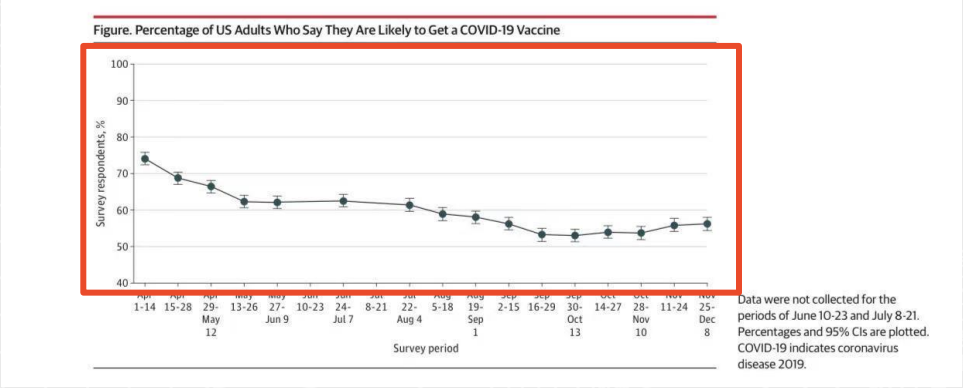
Wouters OJ, Shadlen KC, Salcher-Konrad M, Pollard AJ, Larson HJ, Teerawattananon Y, Jit M. Challenges in ensuring global access to COVID-19 vaccines: production, affordability, allocation, and deployment. The Lancet. 2021 Feb 12.

Olivier J Wouters et al. (2021, *Lancet*)

P=0.06, R2=0.12

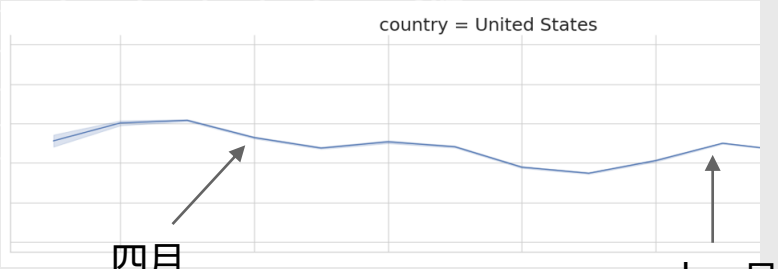
Results

稳健性——与现有问卷研究的比较



Peter G. Szilagyi et al. (2021, *JAMA*)

United States (Twitter)



四月

十一月

本研究的优势

01. 迄今**最大**（106国）的全球新冠疫苗信心比较研究——为被既往研究忽视的数十国家提供疫苗犹豫数据

02. 迄今**唯一**各国/区域疫苗接受度变化的研究，可作为政策制定参考；
迄今唯一关于不信任政府与疫苗谣言的全球定量研究

03. 推特数据丰富；全样本

04. 使用了性能最好的深度学习模型之一，达到了超越人工分析的精度

THANKS!

感谢聆听

Xinyu ZHOU
xinyuzhou17@fudan.edu.cn

CREDITS: This presentation template was created by Slidesgo, including icons by Flaticon, and infographics & images by Freepik

Please keep this slide for attribution